打开网页

浏览器打开 http://localhost:7000（或 https://felicityt1998.github.io/md6329-check-tha/）→ 点左侧讲座 → 按 F 全屏。

此时高亮在 `def main():`。

全程只用 **Shift+→**（下面写作 ⇧→）；只有 State 3 有一次 → 和一次 u。
按多了：**Shift+←** 退回一步。

⇧→ × 3

标题 + 「The problem」段

A 55-year-old sees their GP for the first time with hip pain. Implants now last decades, and younger patients carry more revision risk. So the question is not only: is this OA? It is: **will this hip need replacing within ten years?** That decides how hard to push conservative care, and when to refer.

⇧→ × 2

「The data」段 + 时间线图

We use CHECK: about a thousand Dutch adults aged 45 to 65, followed for ten years. **1,705 hips, 93 replaced: 5.5%.** We use only what exists at the first visit: questionnaires, a hip examination, and the same-day radiograph. The radiologists' grades were scored after year 10, so they are never inputs.

⇧→ × 2

「The approach」段 + 总览图

M0 is a penalised logistic regression on 76 clinical features. M3 is a small head on frozen DenseNet feature maps; it turns each hip crop into one logit. **The final model is M0 plus that logit as one extra column.** Everything is nested cross-validation, grouped by participant.

$$\Uparrow\!\!\rightarrow \times 1$$

灰色 note

Now we follow **one real test hip** through the final model. No number on these pages is typed in.

⇧→ × 2

State 1 标题 + 三句话

State 1. The hip's raw values go through preprocessing fitted on the training fold only. This is where leakage would get in.

⇧→ × 3

右侧出现 hip（第 1 下画面不变，正常）

This is a left hip, **replaced in year 7.** I picked it because the image changed its rank the most. It is the best example, not a typical one.

⇧→ × 3

asym = 15.0，z_asym = 2.68（第 1 下不变）

It rotates **15 degrees more** than the other hip. So on examination it looks like the better hip. The other hip enters only through this baseline difference.

⇧→ × 5

sos_t、z_sos = −0.65，最后是灰色 note

Skewed scales get a Yeo-Johnson transform, with lambda fitted on the training fold. Symptoms are mild: **z is minus 0.65.**

⇧→ × 2

State 2 标题 + 三句话

State 2. The clinical model adds weight times value over 77 columns, then recalibrates. This is the baseline. With 93 events, none of our deep tabular models beat it.

⇧→ × 4

cfg → logit_m0 = −3.60 → p_m0 = 0.013（第 1 下不变）

The inner folds chose strong, mostly ridge regularisation. The logit is minus 3.6. After calibration the risk is **0.013, below the base rate.**

⇧→ × 2

贡献条形图 + 百分位文字

Examination asymmetries pull the risk down; knee pain and age push it up. This hip ranks at the **37th percentile.** It would go home with routine advice.

⇧→ × 2

State 3 标题 + 三句话

State 3: the image head. A frozen DenseNet map, 1,024 channels on a 14 by 14 grid, goes through two residual blocks, attention, and a linear layer. Pooling the map to one vector gave AUPRC **0.12**. Keeping the grid gave **0.49**.

⇧→ × 4

fmap_shape；高亮停在 head_forward 这一行

Here is the input: one by 1,024 by 14 by 14. Let's step inside the head.

→ × 1，再 ⇧→ × 5

进入 head_forward，停在 att = torch.softmax(...)

reduce cuts 1,024 channels to 64. The residual blocks keep the grid. scores gives one number per cell. **Softmax turns them into weights that sum to one.**

u × 1（跳出函数）

回到 main，logit_img = 3.658

The logit is **3.658**: exactly the value the training pipeline saved. And this head never saw this participant.

⇧→ × 1

注意力叠加图

The attention sits on the **upper outer joint margin**, where early hip OA usually starts.

⇧→ × 3

文字 → OARSI 表 → 灰色 note

The head also learned to predict 11 OARSI items, as training targets only. The reader found a **grade-3 femoral osteophyte.** The examination looked fine. The pixels did not.

⇧→ × 2

State 4 标题 + 三句话

State 4: fusion. The image logit enters the same elastic net as one more column. Training hips get a logit from a head that never saw them.

⇧→ × 3

z_img = 4.19，w_img = 0.55（第 1 下不变）

The image z-score is 4.19 and its weight 0.55. **This one term adds about 2.3 to the logit.**

⇧→ × 2

logit_fused = −1.36，p_fused = 0.379

The risk moves from **0.013 to 0.379.**

⇧→ × 2

融合贡献图 + 百分位文字

One bar now outweighs every clinical term. The rank jumps from the **37th to the 95th percentile.** And this hip was replaced in year 7.

⇧→ × 2

State 5 标题 + 三句话

State 5. Now all 1,705 hips. At 5.5% prevalence, a model can rank well and still raise mostly false alarms.

⇧→ × 3

event_rate = 0.055（前 2 下不变）

A random model's AUPRC equals the event rate: **0.055.**

⇧→ × 2

auprc_m0 = 0.383，auprc_fused = 0.586

The clinical model reaches **0.38.** The fusion reaches **0.59.**



max_lift = 18.3，lift_fused = 10.7

The best possible lift is about 18. The fusion reaches **10.7 times chance.**

⇧→ × 2

PR 曲线 + 「A clinic acts on a threshold」

The fusion curve stays above the clinical curve everywhere. But a clinic acts on a threshold, not a curve.

⇧→ × 2

at_m0、at_fused 两个字典

At 10% risk, the fusion flags fewer hips, yet catches **65% of replacements instead of 55%.** Precision rises from 0.21 to 0.37.

⇧→ × 1

灰色 note

These are repeat-0 numbers. The report averages five repeats: **0.37 and 0.55.**

⇧→ × 2

结论表

To sum up. Clinical model: AUPRC 0.37. Adding the pixel logit: **0.55**, and the gain holds without early surgeries. Across centres, ranking transports; calibration does not.

⇧→ × 2

结尾段 + 参考文献

So a first-visit score from routine data and the raw film could guide review and referral. Two cautions: the film partly encodes the surgeon's own decision, and calibration must be redone locally. Next: external validation. **Thank you.**